

$(\mathbb{R}^n, \mu)$, $(\Omega, \mathcal{F}, \mathbb{P})$. A white noise on $\mathbb{R}^n$ is a map $\dot{W}$ that assigns every finite Borel set A a Gaussian rv $\dot{W}(A)$ with μ Lebesgue.

$$(1) \dot{W}(A) \sim \mathcal{N}(0, \mu(A))$$

$$(2) E(\dot{W}(A)\dot{W}(B)) = \mu(A \cap B)$$

$\Rightarrow$ if $A \cap B = \emptyset$, then

• $\dot{W}(A), \dot{W}(B)$ independent

$$\dot{W}(A \cup B) = \dot{W}(A) + \dot{W}(B)$$

The covariance map in (2) is positive definite, ie for any sets $A_1, \dots, A_n$, reals $a_1, \dots, a_n$, we have

$$\sum_{i,j} E(\dot{W}(A_i)\dot{W}(A_j)) a_i a_j \geq 0$$

$$\begin{aligned} \text{Pf. } \sum_{i,j} \mu(A_i \cap A_j) a_i a_j &= \sum_{i,j} \int_{\mathbb{R}} a_i a_j \mathbb{1}_{A_i}(x) \mathbb{1}_{A_j}(x) dx \\ &= \int_{\mathbb{R}} \underbrace{\sum_{i,j} a_i a_j \mathbb{1}_{A_i}(x) \mathbb{1}_{A_j}(x)}_{\left(\sum_i a_i \mathbb{1}_{A_i}(x)\right)^2} dx \geq 0 \quad \square \end{aligned}$$

Does white noise exist?

Def A stochastic process $X = (X_t, t \in T)$ is a real-valued Gaussian process if for all choices $t_1, \dots, t_n \in T$ the vector $(X_{t_1}, \dots, X_{t_n})$ is an $\mathbb{R}^n$ -valued Gaussian random vector.

Thm Given any index set T , function $\mu: T \rightarrow \mathbb{R}$, and symmetric, positive definite function $\Sigma: T \times T \rightarrow \mathbb{R}$, $\exists$ Gaussian process $X = (X_t: t \in T)$ with mean μ , cov Σ . $\Rightarrow$ existence of $\dot{W}$.

$A \cap B = \emptyset \Rightarrow \dot{W}(A \cup B) = \dot{W}(A) + \dot{W}(B)$, a.s. In fact, for $A_1, A_2, \dots$ disjoint and $\sum_{i=1}^{\infty} \mu(A_i) < \infty$,

$$\dot{W}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \dot{W}(A_i), \text{ a.s. (induct, take limit) to show for } \infty$$

Is $\dot{W}$ a σ -finite signed measure $\forall \omega$? No, $\sum_{i=1}^{\infty} \dot{W}(A_i)$ is not absolutely convergent when $\dot{W}(\bigcup A_i)$ is finite, hence ill-defined wrt rearranging terms.

Exercise!!

However, $\dot{W}$ is a σ -finite $L^2(\Omega)$ -valued countably additive measure.

$\downarrow \hookrightarrow \mathcal{B}(\mathbb{R})$ $\hookrightarrow \mathcal{B}(\mathbb{R}^n)$
 $\|\dot{W}(A)\|_2 := \left[E(\dot{W}(A)^2) \right]^{\frac{1}{2}} < \infty, \forall A \in \mathcal{B}(\mathbb{R})$
 $\exists \{A_i\}$ an increasing sequence such that
 $\mathbb{R}^n = \bigcup_{i=1}^{\infty} A_i$ and $\sup_i E(\dot{W}(A_i)^2) < \infty$.

$\forall \{A_i\}$ decreasing sequences with $\bigcap_{i=1}^{\infty} A_i = \emptyset$ and $\mu(A_1) < \infty$,
 $\mathcal{B}(\mathbb{R}^n)$

$\lim_{j \rightarrow \infty} \|\dot{W}(A_j)\|_2 = 0$. (continuity from above?)
 $\uparrow$ follows from continuity of μ .

$\dot{W}(\emptyset) \equiv 0$. ~~not S_0 ?~~ ~~meanst $E(\dot{W}(\emptyset)^2) = 0$?~~
 $\uparrow$ Gaussian with 0 variance has distribution S_0 .

$\dot{W}$ on $\mathbb{R}$ is the weak derivative of Brownian motion.

$W_t = \dot{W}((0, t])$, $t \in \mathbb{R}_+$

defines a Brownian motion (up to continuity issues).

$W_t \sim N(0, \mu(0, t) = t)$,

$E(W_t W_s) = E(\dot{W}((0, t]) \dot{W}((0, s])) = \mu([0, t] \cap (0, s]) = t \wedge s$.

Likewise, given $\dot{W}$ on $\mathbb{R}^n$ and $t = (t_1, \dots, t_n) \in \mathbb{R}_+^n$,

we can define the n -dim Brownian sheet $\leftarrow$ Gaussian field

$W_t = \dot{W}((0, t_1] \times \dots \times (0, t_n])$.

To define a space-time white noise $\dot{W}(dt, dx)$ on $\mathbb{R}_+ \times \mathbb{R}^n$ we take a white noise $\dot{W}$ on $\mathbb{R}^{n+1}$ restricted to subsets of $\mathbb{R}_+ \times \mathbb{R}^n$.

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